January 10th, 2026

To: XYZ Energy

Subject: Structural Certification for Installation of Solar Panels

To whom it may concern:

A design check for the subject building was done on the existing roofing and framing systems for the installation of solar panels over the roof. From a site visit conducted by myself and the as-built structural drawings for the newer addition, the existing roof support structures were observed as follows:

Roof A

The roof structure consists of rubber membrane over rigid insulation that is supported by type A, 1.5-in, 20-gauge corrugated steel deck. The steel deck is supported by 32LA12 steel open web joists at 5'-6" o.c.

spanning in northwest-southeast direction. There are 7 bays of steel joists, each with a bay width of 52'-0". The 32LA12 steel joists are supported by W18x50 steel wide flange I-girders, spanning perpendicular to the joists. There are 17 bays of W18x50 girders, each with a bay width of 22'-0". There are shear stiffners added to the W18x50 girders under each 32LA12 joists. One of the 22'-0" bays is framed with W12x22 steel joists at 48" o.c.. The roof has a close to flat slope.

Roofs B and C

The roof structure consists of standing seam metal deck over rigid insulation on type B, 1.5", 22-gauge corrugated steel deck. The steel deck is supported by 30-in pre-fabricated steel open web joists at 60" o.c. at maximum, spanning in northwest-southeast direction. There are 5 bays of steel joists, with bay widths of 27'-9", 40'-6", 40'-6", 40'-6", and 28'-6", from north to south. The steel joists are simply supported by steel rigid frames, spanning in perpendicular direction to the joists. The steel rigid frames have an out-to-out width of 366'-0", and a 3-degree slope. The rigid frames are supported by steel columns at 45'-6' to 46'-6" o.c..

According to IEBC 2021, 805.2 Addition or replacement of roofing or replacement of equipment.

"Any existing gravity load-carrying structural element for which an alteration causes an increase in design dead, live or snow load, including snow drift effects, of more than 5 percent shall be replaced or altered as needed to carry the gravity loads required by the International Building Code for new structures. Any existing gravity load-carrying structural element whose gravity load-carrying capacity is decreased as part of the alteration shall be shown to have the capacity to resist the applicable design dead, live and snow loads, including snow drift effects, required by the International Building Code for new structures."

The roof structure was then re-analyzed by applying IBC 2021 code and standards, with Massachusetts Building Code 10th Edition to evaluate the structural integrity under the impact of solar installation.

The existing roof framing systems are judged to be adequate to withstand the loadings imposed by the installation of the solar panels. No reinforcement is necessary. The roof can support another 10.0 psf of solar ballast dead load on roof A, and another 4.0 psf of solar flushmount dead load on roofs B and C, on top of existing dead and snow loads. Please note that the maximum allowable ballast blocks per tray is 3.

The following are tables of the solar standoffs spacing per on Roofs B and C:

Roofs B and C	Solar Standoffs that Connect to the Standing Seam Metal Deck		
	Zone 1	Zone 2	Zone 3
Exposed	48" o.c. with a staggered pattern	24" o.c. with a staggered pattern	24" o.c. with a staggered pattern
Not Exposed	48" o.c. with a staggered pattern	48" o.c. with a staggered pattern	24" o.c. with a staggered pattern

I further certify that all applicable loads required by IBC 2021 with Massachusetts amendments were applied to the solar rail system and analyzed. The applicable loads are indicated in the load combination table and the summation of maximum loads indicated on permit plan, which is part of this submittal package. Furthermore, I was informed that the installation crews have been trained to install the solar panels based on the specific roof installation instructions developed by the manufacturer for the solar racking system. Finally, I accept the certifications indicated by the solar panel manufacturer for the ability of the panels to withstand high wind and snow loads.

Sincerely,

XYZ
Structural Engineer

Load and Load Combination:

Dead Load: Rubber Membrane

Rubber Membrane + Coverboard = 2.5 psf
Misc (Insulation, Vapor Barrier, etc) = 5.0 psf
Corrugated Steel Deck = 2.5 psf
Hung Loads (Mechanical, Electrical, Plumbing) = 5.0 psf
Solar Panels = 10.0 psf (Given by Product Catalog)
Total Dead Load = 2.5 + 5.0 + 2.5 + 5.0 + 10.0 = **25.0 psf**

Dead Load: Standing Seam Roof

Standing Seam Metal Roof = 2.5 psf
Misc (Insulation, Vapor Barrier, etc) = 2.0 psf
Corrugated Steel Deck = 2.5 psf
Hung Loads (Mechanical, Electrical, Plumbing) = 5.0 psf
Solar Panels = 5.0 psf (Given by Product Catalog)
Total Dead Load = 2.5 + 2.0 + 2.5 + 5.0 + 5.0 = **17.0 psf**

Snow Load:

Ground Snow Load, P_g:
City of Leominster, MA = 60 psf (Mass Building Code, 780 CMR, 10th Ed.)

Flat Roof snow load = $P_f = 0.7\ C_e C_t I\ p_g$ = 0.7 x 1.0 x 1.0 x 1.0 x 60 psf = 42.0 psf (ASCE-7, Eq7.3-1)

C_e = exposure factor = 1.0, (ASCE-7, Table 7.3-1), Terrain B, partially exposed
C_t = thermal factor = 1.0 (ASCE-7, Table 7.3-2),
I = importance factor = 1.0 (ASCE-7, Table 1.5-2), occupancy category II

Sloped roof snow load $P_s = C_s\ P_f$ = 1.0 x P_f = 42.0 psf (ASCE-7, Equation 7.4-1)

C_s = slope factor = 1.0 (ASCE-7 Fig. 7.4-1), Cold roofs with C_t = 1.0

⇨ Use **42.0 psf** roof snow load per Mass Building Code 10th Ed.

Wind Speed:

Basic Wind Speed:
City of Leominster, MA = 115 mph (Mass Building Code, 780 CMR, 10th Ed.)

Wind Pressure:

Q_z = 0.00256 K_z K_{zt} K_d K_e V^2 (ASCE-7 2016 Eq. 26.10-1)

K_Z = 0.76 (40' building) (ASCE-7 2016 Table 26.10-1)
K_{zt} = 1.0 (ASCE-7 2016 26.8.2)
K_d = 0.85 (ASCE-7 2016 Table 26.6-1)
K_e = 1.0 (ASCE-7 2016 26.9)
V_{utl} = 115 mph (IBC 2018)

Therefore, q_z = 0.00256 x 0.76 x 1.0 x 0.85 x 1.0 x 115^2 = **21.87 psf**

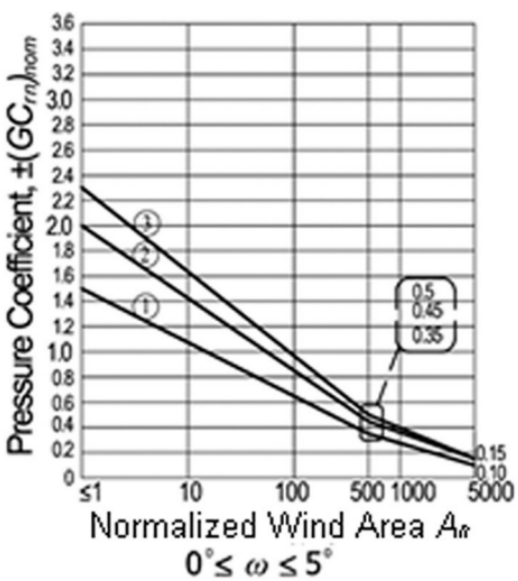
<u>Solar Ballast System Wind Pressure (uplifting): (ASCE-7 2016 Eq. 29.4-5)</u>

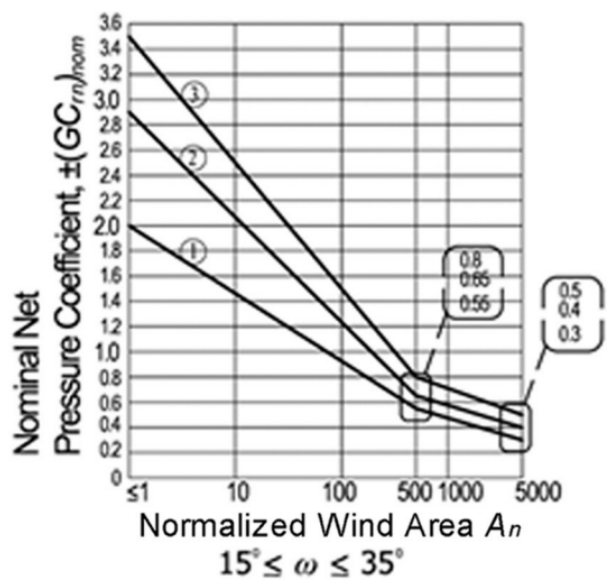
P = q_h (GC_{rn}) = 21.87 psf x -1.680 = -36.74 psf (zone 1)
P = q_h (GC_{rn}) = 21.87 psf x -2.400 = -52.49 psf (zone 2)
P = q_h (GC_{rn}) = 21.87 psf x -2.880 = -62.99 psf (zone 3)

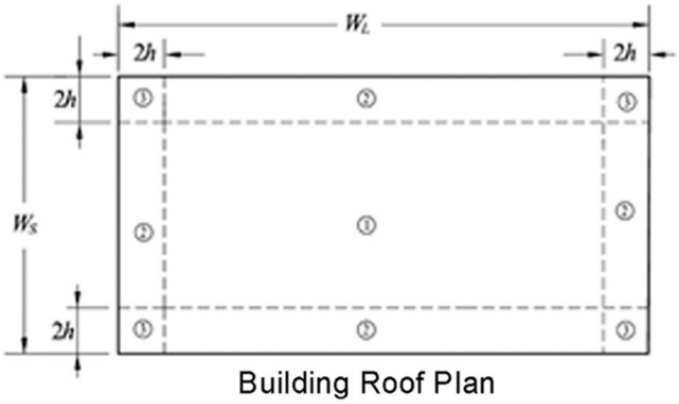
GC_{rn} = (γ_p)(γ_c) (γ_E)(GC_{rn})$_{nom}$ (ASCE-7 2016 Eq. 29.4-6)
= (1.0) (0.8) (1.5) (1.40) = 1.680 (wind zone 1)
= (1.0) (0.8) (1.5) (2.00) = 2.400 (wind zone 2)
= (1.0) (0.8) (1.5) (2.40) = 2.880 (wind zone 3)

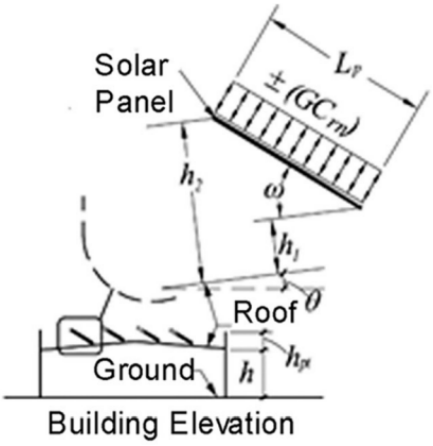
(GC_{rn})$_{nom}$ = 1.40 (zone1), 2.00 (zone 2), 2.40 (zone 3) for normalized area of 15.3 ft^2
γ_P = 1.0 to be conservative (ASCE-7 2016 29.4.3)
γ_C = 0.8 maximum (ASCE -7 2016 29.4.3)
γ_E = 1.5 for panels within 10' around the edge, or 1.0 for interior (ASCE-7 2016 29.4.4)

a_{pv} = 0.4 (h x W_L)$^{0.5}$ = 0.4 x (40 ft x 375 ft)$^{0.5}$ = 48.98 ft use 40.0 ft
normalized wind area = (1000/a_{pv}^2) x A = (1000/1600) x (24.5 ft^2) = 15.3 ft^2









<u>Solar Flushmount Wind Pressure:</u>
$P = q_h (GC_p) (\gamma_E)(\gamma_a)$ = 21.87 psf x -1.6 x 1.5 x 0.75 = -39.37 psf (uplifting zone 1)
$P = q_h (GC_p) (\gamma_E)(\gamma_a)$ = 21.87 psf x -2.1 x 1.5 x 0.75 = -51.67 psf (uplifting zone 2)
$P = q_h (GC_p) (\gamma_E)(\gamma_a)$ = 21.87 psf x -2.8 x 1.5 x 0.75 = -68.89 psf (uplifting zone 3)

GC_p = -1.6 (zone 1), -2.1 (zone 2), -2.8 (zone 3)
γ_E = 1.5 for edge panels, or 1.0 for interior (ASCE-7 2016 29.4.4)
γ_a = 0.75 for wind effective area of 20 ft^2 (Figure 29.4-8, ASCE-7 2016)

$P = q_h (GC_p) (\gamma_E)(\gamma_a)$ = 21.87 psf x 0.30 x 1.5 x 0.75 = 7.38 psf (downward) use 16.0 psf

GC_p = 0.30 (all zones)
γ_E = 1.5 for edge panels, or 1.0 for interior (ASCE-7 2016 29.4.4)
γ_a = 0.75 for wind effective area of 20 ft^2 (Figure 29.4-8, ASCE-7 2016)

<u>Seismic Load</u>

S_S = 0.284 (Mass Building Code 10th ed. Table 1604.11)
S_1 = 0.068 (Mass Building Code 10th ed. Table 1604.11)

Spectral Response Acceleration coefficients:

$S_{MS} = F_a S_S$ = 1.6 x 0.284 = 0.454 (IBC Equation 16-20)
$S_{M1} = F_v S_1$ = 2.4 x 0.068 = 0.163 (IBC Equation 16-21)

Design Spectral Response Acceleration Parameters:

S_{DS} = 2/3 x S_{MS} = 2/3 x 0.454 = 0.303 (IBC Equation 16-22)

S_{D1} = 2/3 x S_{M1} = 2/3 x 0.163 = 0.109 (IBC Equation 16-23)

According to IBC Table 1613.3.5 (1) and (2), the seismic design category is "B" and the calculation can be finished by using ASCE-7 equivalent force procedure.

The importance factor, I, for this building is 1.00.

The response modification coefficient, R, is 3.00 according to ASCE-7.

T_L = 6 sec (ASCE Fig 22-14)

$T = C_t h_n^x = 0.028 \times 40^{0.8} = 0.536$ sec (ASCE-7 Equation 12.8-7)

The seismic response coefficient, C_s, is

$C_s = S_{DS} I/R = 0.303 \times 1.00/3.00 = 0.101$

The maximum value of $C_s = S_{D1}I/(TR) = 0.109 \times 1.00/(0.536 \times 3.00) = 0.0677$

The minimum value of C_s = 0.01

Therefore, C_s = 0.0677

The horizontal base shear force resulted from C_s is significantly less than the horizontal wind forces. Therefore, seismic design should not be considered.

Load Combination (ASD): ASCE 7-16 Chapter 2.4 – Rubber Membrane

D + S = **67.0 psf**
D + 0.75 (0.6W) + 0.75S = 63.70 psf
0.6D + 0.6W = -22.80 psf

Load Combination (ASD): ASCE 7-16 Chapter 2.4 – Standing Seam Metal

D + S = **59.0 psf**
D + 0.75 (0.6W) + 0.75S = 55.70 psf
0.6D + 0.6W = -27.59 psf

Roof Structures

Roof A

The roof structure consists of rubber membrane over rigid insulation that is supported by type A, 1.5-in, 20-gauge corrugated steel deck. The steel deck is supported by 32LA12 steel open web joists at 5'-6" o.c., spanning in northwest-southeast direction. There are 7 bays of steel joists, each with a bay width of 52'-0". The 32LA12 steel joists are supported by W18x50 steel wide flange I-girders, spanning perpendicular to the joists. There are 17 bays of W18x50 girders, each with a bay width of 22'-0". There are shear stiffners added to the W18x50 girders under each 32LA12 joists. One of the 22'-0" bays is framed with W12x22 steel joists at 48" o.c.. The roof has a close to flat slope.

Roofs B and C

The roof structure consists of standing seam metal deck over rigid insulation on type B, 1.5", 22-gauge corrugated steel deck. The steel deck is supported by 30-in pre-fabricated steel open web joists at 60" o.c. at maximum, spanning in northwest-southeast direction. There are 5 bays of steel joists, with bay widths of 27'-9", 40'-6", 40'-6", 40'-6", and 28'-6", from north to south. The steel joists are simply supported by steel rigid frames, spanning in perpendicular direction to the joists. The steel rigid frames have an out-to-out width of 366'-0", and a 3-degree slope. The rigid frames are supported by steel columns at 45'-6' to 46'-6" o.c..

Structural Analysis and Calculation

Roof A

According to Vulcraft product catalog, the maximum total uniform load that the deck can withstand is approximately 70 psf for 5’-6” with three-span. Therefore, the 1.5” type B deck is adequate to support the added PV load with a total roof load combination of 67.0 psf with unsupported length of 5’-6”.

VERTICAL LOADS TYPE 1.5 A

Span Condition	Type No.	Max. Span SDI Const. Load		Allowable Total (Dead + Live) Uniform Load (Lbs./Sq. Ft.)														
				Span (ft.-in.)														
		Stress (26.6 KSI)	Δ = L/240	4 -0	4 -6	5 -0	5 -6	6 -0	6 -6	7 -0	7 -6	8 -0	8 -6	9 -0	9 -6	10 -0	10 -6	11 -0
ONE	1.5A22	4’-2	4’-6	79	63	51	42	35	30	26	23	20						
	1.5A21	4’-9	4’-10	90	71	58	48	40	34	29	26	22	20					
	1.5A20	5’-3	5’-2	100	79	64	53	44	38	33	28	25	22	20				
	1.5A19	6’-3	5’-8	118	93	76	63	53	45	39	34	30	26	23	21			
	1.5A18	7’-2	6’-2	136	107	87	72	60	51	44	39	34	30	27	24	22	20	
TWO	1.5A22	5’-2	5’-6	87	68	55	46	39	33	28	25	22						
	1.5A21	5’-10	5’-11	98	77	62	52	43	37	32	28	24	22					
	1.5A20	6’-5	6’-2	106	84	68	56	47	40	35	30	26	23	21				
	1.5A19	7’-8	6’-9	123	97	79	65	55	47	40	35	31	27	24	22	20		
	1.5A18	8’-10	7’-3	140	111	90	74	62	53	46	40	35	31	28	25	22	20	
THREE OR MORE	1.5A22	5’-2	5’-6	108	86	69	57	48	41	35	31	27	24	21				
	1.5A21	5’-10	5’-11	122	96	78	[illegible]	54	46	40	35	30	27	24	22	20		
	1.5A20	6’-5	6’-2	132	105	85	70	59	50	43	38	33	29	26	23	21		
	1.5A19	7’-8	6’-9	154	122	99	[illegible]	69	58	50	44	39	34	30	27	25	22	20
	1.5A18	8’-10	7’-3	175	138	112	93	78	66	57	50	44	39	35	31	28	25	23

32LA12 Steel Joists

Member Size = 32LA12

Spacing = 66” O.C.

Span Length = 52’-0” approximately

w = 67.0 psf x 66 in/(12 in/ft) + 31.0 lbs/ft (joist selfweight) = 399.5 lbs/ft

According to the SJI chart, the joists' capacity when the span is 52'-0" is approximately 439 plf total, which is greater than 399.5 plf from demand. Therefore, the steel joists are structurally adequate to carry the total loads.

<u>W18x50</u>

<u>Check moment capacity:</u>

W = 67.0 psf x 52.0' = 3484 lbs/ft = 3.484 K/ft say 4.00 K/ft

$M = 0.078wL^2$ = 0.078 x 4.00 K/ft (selfweight included) x 22.0^2 ft^2 = 151 K-ft

M_{px}/Ω_b = 252 K-ft x 0.66 = 166 K-ft, OK

<u>Check shear capacity:</u>

V = 4.00 K/ft x 22.0 ft/2 = 44.0 K

V_{nx}/Ω_v = 128 K x 0.66 = 84.50 K, OK

<u>Check deflection:</u>

$\Delta = 0.0065wL^4/EI$ = [0.0065 x 4.00 K/ft x 22.0^4 ft^4 x 12^3 in^3/ft^3]/[(29000 Ksi)(800 in^4)] = 0.454 in

L/360 = 22.0' x 12 in/ft/360 = 0.733 in, OK

W10x22

Check moment capacity:

W = 67.0 psf x 4.0' = 268 lbs/ft = 0.268 K/ft say 0.500 K/ft

$M = 0.125wL^2$ = 0.125 x 0.500 K/ft (selfweight included) x 22.0^2 ft^2 = 30.25 K-ft

M_{px}/Ω_b = 64.9 K-ft x 0.66 = 42.83 K-ft, OK

Check shear capacity:

V = 0.500 K/ft x 22.0 ft/2 = 5.50 K

V_{nx}/Ω_v = 48.8 K x 0.66 = 32.21 K, OK

Check deflection:

$\Delta = 5wL^4/384EI$ = [5 x 0.500 K/ft x 22.0^4 ft^4 x 12^3 in^3/ft^3]/[384(29000 Ksi)(118 in^4)] = 0.770 in

L/240 = 22.0' x 12 in/ft/240 = 1.100 in, OK

W12x58

Check moment capacity:

W = 67.0 psf x 11.0' = 737 lbs/ft = 0.737 K/ft say 1.00 K/ft

$M = 0.125wL^2$ = 0.125 x 1.00 K/ft (selfweight included) x 26.0^2 ft^2 = 84.50 K-ft

M_{px}/Ω_b = 216 K-ft x 0.66 = 142.56 K-ft, OK

Check shear capacity:

V = 1.00 K/ft x 26.0 ft/2 = 13.0 K

V_{nx}/Ω_v = 87.8 K x 0.66 = 57.95 K, OK

Check deflection:

$\Delta = 5wL^4/384EI = [5 \times 1.00\ K/ft \times 26.0^4\ ft^4 \times 12^3\ in^3/ft^3]/[384(29000\ Ksi)(475\ in^4)] = 0.746\ in$

$L/360 = 26.0' \times 12\ in/ft/360 = 0.833\ in$, OK

Roofs B and C

According to Vulcraft product catalog, the maximum total uniform load that the deck can withstand is approximately 124 psf for 5’-0” with three-span. Therefore, the 1.5” type B deck is adequate to support the added PV load with a total roof load combination of 59.0 psf with unsupported length of 5’-0”.

VERTICAL LOADS FOR TYPE 1.5B

Batts are field installed and may require separation.

No. of Spans	Deck Type	Max. SDI Const. Span	Allowable Total (PSF) / Load Causing Deflection of L/240 or 1 inch (PSF)										
			Span (ft.-in.) ctr to ctr of supports										
			5-0	5-6	6-0	6-6	7-0	7-6	8-0	8-6	9-0	9-6	10-0
1	B24	4'-8	115 / 56	95 / 42	80 / 32	68 / 26	59 / 20	51 / 17	45 / 14	40 / 11	35 / 10	32 / 8	29 / 7
	B22	5'-7	98 / 81	81 / 61	68 / 47	58 / 37	50 / 30	44 / 24	38 / 20	34 / 17	30 / 14	27 / 12	25 / 10
	B20	6'-5	123 / 105	102 / 79	86 / 61	73 / 48	63 / 38	55 / 31	48 / 26	43 / 21	38 / 18	34 / 15	31 / 13
	B19	7'-1	146 / 129	121 / 97	101 / 75	86 / 59	74 / 47	65 / 38	57 / 31	51 / 26	45 / 22	40 / 19	36 / 16
	B18	7'-8	168 / 152	138 / 114	116 / 88	99 / 69	85 / 55	74 / 45	65 / 37	58 / 31	52 / 26	46 / 22	42 / 19
	B16	8'-8	215 / 196	178 / 147	149 / 113	127 / 89	110 / 71	96 / 58	84 / 48	74 / 40	66 / 34	60 / 29	54 / 24
2	B24	5'-10	124 / 153	103 / 115	86 / 88	74 / 70	64 / 56	56 / 45	49 / 37	43 / 31	39 / 26	35 / 22	31 / 19
	B22	6'-11	100 / 213	83 / 160	70 / 124	59 / 97	51 / 78	45 / 63	39 / 52	35 / 43	31 / 37	28 / 31	25 / 27
	B20	7'-9	128 / 267	106 / 201	89 / 155	76 / 122	66 / 97	57 / 79	51 / 65	45 / 54	40 / 46	36 / 39	32 / 33
	B19	8'-5	150 / 320	124 / 240	104 / 185	89 / 145	77 / 116	67 / 95	59 / 78	52 / 65	47 / 55	42 / 47	38 / 40
	B18	9'-1	169 / 369	140 / 277	118 / 213	101 / 168	87 / 134	76 / 109	67 / 90	59 / 75	53 / 63	48 / 54	43 / 46
	B16	10'-3	213 / 471	176 / 354	149 / 273	127 / 214	110 / 172	95 / 140	84 / 115	74 / 96	66 / 81	60 / 69	54 / 59
3	B24	5'-10	154 / 120	128 / 90	108 / 69	92 / 55	79 / 44	69 / 35	61 / 29	54 / 24	48 / 21	43 / 17	39 / 15
	B22	6'-11	124 / 167	103 / 126	87 / 97	74 / 76	64 / 61	56 / 50	49 / 41	43 / 34	39 / 29	35 / 24	31 / 21
	B20	7'-9	159 / 209	132 / 157	111 / 121	95 / 95	82 / 76	72 / 62	63 / 51	56 / 43	50 / 36	45 / 31	40 / 26
	B19	8'-5	186 / 250	154 / 188	130 / 145	111 / 114	96 / 91	84 / 74	74 / 61	65 / 51	58 / 43	52 / 37	47 / 31
	B18	9'-1	210 / 289	174 / 217	147 / 167	126 / 132	108 / 105	95 / 86	83 / 71	74 / 59	66 / 50	59 / 42	54 / 36
	B16	10'-3	264 / 369	219 / 277	185 / 214	158 / 168	136 / 135	119 / 109	105 / 90	93 / 75	83 / 63	74 / 54	67 / 46

The following is a screenshot of the as-built Butler Manufacturing structural drawing, dated 02/07/2020. Please note that there is 5.0 psf collateral dead load incorporated in the roof framing design. The solar dead load is estimated to be less than 4.0 psf. Therefore, the roof structure can adequately support the solar array.

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Codes and Loads
WHEN MULTIPLE BUILDINGS ARE INVOLVED, SPECIFIC LOAD FACTORS FOR DIFFERING OCCUPANCIES, BUILDING DIMENS
HEIGHTS, FRAMING SYSTEMS, ROOF SLOPES, ETC., MAY RESULT IN DIFFERENT LOAD APPLICATION FACTORS THAN
INDICATED BELOW.  SEE CALCULATIONS FOR FURTHER DETAILS.  WIND LOADS ARE APPLIED TO OVERALL BUILDING EN
COMMON WALLS BETWEEN CONNECTED SHAPES ARE NOT SUBJECT TO EXTERNAL WIND LOADS.
City: Leominster                    County: Worcester                    State: Massachusetts

Building Code
Building Code: Massachusetts State Building Code (9th Ed.)                          Structural:
Building Risk/Occupancy Category: II (Standard Occupancy Structure)                 Cold Form: 1


Dead and Collateral Loads                        Material Dead Weight
Collateral Gravity:5.00 psf                      Roof Covering + Second. Dead Load: Varies
Collateral Uplift:  0.00 psf                     Frame Weight (for seismic):3.25 psf

Wind Load                                        Snow Load
Wind Speed: Vult= 121.00 mph (Vasd = 93.73mph)   Ground Snow Load: pg: 60.00 psf
The 'Envelope Procedure' is Used - User Modified Flat Roof Snow: pf: 42.00 psf
Wind Exposure: B - Kz: 0.761                     Design Snow (Sloped): ps: 42.00 psf
Parts Wind Exposure Factor: 0.761                Rain Surcharge: 0.00
Wind Enclosure: Enclosed                         Specified Minimum Roof Snow: 35.00 psf
Topographic Factor: Kzt: 1.0000                  Exposure Factor: 2 Partially Exposed - Ce: 1.00
                                                 Snow Importance: Is: 1.000
NOT Windborne Debris Region                      Thermal Factor: Heated - Ct: 1.00
Base Elevation: 4/0/0                            Ground / Roof Conversion: 0.70
Primary Zone Strip Width: 2a: 29/0/0             Obstructed or Not Slippery
Parts / Portions Zone Strip Width: a: 14/6/0
Basic Wind Pressure: q: 24.24 psf
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Roof Attachments

$Q_z = 0.00256\ K_z\ K_{zt}\ K_d\ K_e\ V^2$ (ASCE-7 2016 Eq. 26.10-1)

K_Z = 0.76 (40' building) (ASCE-7 2016 Table 26.10-1)
K_{zt} = 1.0 (ASCE-7 2016 26.8.2)
K_d = 0.85 (ASCE-7 2016 Table 26.6-1)
K_e = 1.0 (ASCE-7 2016 26.9)
V_{utl} = 115 mph (IBC 2018)

Therefore, q_z = 0.00256 x 0.76 x 1.0 x 0.85 x 1.0 x 115^2 = **21.87 psf**

Solar Flushmount Wind Pressure:

$P = q_h\ (GC_p)\ (\gamma_E)(\gamma_a)$ = 21.87 psf x -1.6 x 1.5 x 0.75 = -39.37 psf (uplifting zone 1)
$P = q_h\ (GC_p)\ (\gamma_E)(\gamma_a)$ = 21.87 psf x -2.1 x 1.5 x 0.75 = -51.67 psf (uplifting zone 2)
$P = q_h\ (GC_p)\ (\gamma_E)(\gamma_a)$ = 21.87 psf x -2.8 x 1.5 x 0.75 = -68.89 psf (uplifting zone 3)

GC_p = -1.6 (zone 1), -2.1 (zone 2), -2.8 (zone 3)
γ_E = 1.5 for edge panels, or 1.0 for interior (ASCE-7 2016 29.4.4)
γ_a = 0.75 for wind effective area of 20 ft^2 (Figure 29.4-8, ASCE-7 2016)

The tributary area of each standoff is = 7.00 ft x 4.00 ft /2 = 14.0 ft^2

The uplifting wind forces are then

F_1 = -39.37 psf x 14.00 ft^2 = -551 lbs (zone 1)
F_2 = -51.67 psf x 14.00 ft^2 = -723 lbs (zone 2)
F_3 = -68.89 psf x 14.00 ft^2 = -964 lbs (zone 3)

The adjusted uplifting wind forces after IBC 2021 load combination 16-15.

F_1 = -551 lbs x 0.6 = -331 lbs, OK (zone 1)
F_2 = -723 lbs x 0.6 = -434 lbs, OK (zone 2)
F_3 = -964 lbs x 0.6 = -579 lbs, OK (zone 3)

Therefore, the spacing of the solar standoffs will be 48" oc/24" oc/24" oc for wind zones 1/2/3, respectively.